

# Linear Algebra

## Lecture 16

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## Examples

(1) Let  $\mathbf{A} \in \mathbb{C}^{3 \times 3}$  be such that  $\mathbf{A}^2 = 6\mathbf{A}$  and  $\text{trace } \mathbf{A} = 12$ . Let us determine the eigenvalues of  $\mathbf{A}$ .

Consider the polynomial  $p(t) = t^2 - 6t$ . Since  $p(\mathbf{A}) = \mathbf{O}$ , we see that  $p(\lambda) = \lambda^2 - 6\lambda = \lambda(\lambda - 6) = 0$  for every eigenvalue  $\lambda$  of  $\mathbf{A}$ . Since  $\text{trace } \mathbf{A} = 12$ , the sum of the eigenvalues of  $\mathbf{A}$  (counting algebraic multiplicities) is equal to 12. Hence the eigenvalues of  $\mathbf{A}$  are 6, 6, 0, that is, 6 is an eigenvalue of  $\mathbf{A}$  of algebraic multiplicity 2, and 0 is an eigenvalue of  $\mathbf{A}$  of algebraic multiplicity 1.

(ii) Let  $\mathbf{A} := [\mathbf{e}_2 \ \mathbf{e}_3 \ \cdots \ \mathbf{e}_n \ \mathbf{e}_1]$ , where  $\mathbf{e}_1, \dots, \mathbf{e}_n$  are the basic column vectors in  $\mathbb{C}^{n \times 1}$ .

Then  $\mathbf{A}\mathbf{e}_1 = \mathbf{e}_2, \dots, \mathbf{A}\mathbf{e}_{n-1} = \mathbf{e}_n$  and  $\mathbf{A}\mathbf{e}_n = \mathbf{e}_1$ . Hence

$$\mathbf{A} \begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^T = \begin{bmatrix} x_n & x_1 & x_2 & \cdots & x_{n-1} \end{bmatrix}^T$$
for  $\begin{bmatrix} x_1 & x_2 & \cdots & x_{n-1} & x_n \end{bmatrix}^T \in \mathbb{C}^{n \times 1}$ .

The matrix **A** represents a **cyclic shift to the right**. Observe that  $\mathbf{A}^n = \mathbf{I}$  and also that **A** is a unitary matrix given by

$$\mathbf{A} := \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & \cdots & 0 \\ \vdots & \vdots & \cdots & \cdots & \cdots & \cdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Let  $\lambda$  be an eigenvalue of **A**. Since  $\mathbf{A}^n = \mathbf{I}$ , we see that  $\lambda^n = 1$ . Let  $\omega := e^{2\pi i/n}$ . Then the  $n$ th roots of 1 are  $1, \omega, \omega^2, \dots, \omega^{n-1}$ . Thus  $\lambda \in \{1, \omega, \omega^2, \dots, \omega^{n-1}\}$ .

Conversely, we show that each  $1, \omega, \omega^2, \dots, \omega^{n-1}$  is an eigenvalue of **A** by finding a corresponding eigenvector.

Let  $\mathbf{x} := [x_1 \ x_2 \ \cdots \ x_{n-1} \ x_n]^T \in \mathbb{C}^{n \times 1}$ . Then  $\mathbf{A}\mathbf{x} = \lambda \mathbf{x}$  means  $x_1 = \lambda x_2, x_2 = \lambda x_3, \dots, x_{n-1} = \lambda x_n$  and  $x_n = \lambda x_1$ .

For  $j = 0, \dots, n-1$ , define

$$\mathbf{x}_j := [1 \ 1/\omega^j \ 1/\omega^{2j} \ \cdots \ 1/\omega^{(n-1)j}]^T.$$

Now  $1/\omega^{(n-1)j} = \omega^j$  since  $\omega^n = 1$ , and so

$$\mathbf{x}_j := [1 \ 1/\omega^j \ 1/\omega^{2j} \ \cdots \ 1/\omega^{(n-2)j} \ \omega^j]^T.$$

Hence  $\mathbf{A}\mathbf{x}_j = [\omega^j \ 1 \ 1/\omega^j \ 1/\omega^{2j} \ \cdots \ 1/\omega^{(n-2)j}]^T = \omega^j \mathbf{x}_j$ . Thus  $\mathbf{x}_j$  is an eigenvector of  $\mathbf{A}$  corresponding to the eigenvalue  $\omega^j$  for  $j = 0, 1, \dots, n-1$ .

**Remark:**  $\mathbf{A}$  is an example of a **circulant matrix**. In general a **circulant matrix** is a square matrix each of whose rows are obtained by a cyclic shift to the right of the preceding rows.

# Unitarily Diagonalizable Matrix

In the last lecture we saw that if the scalars are complex numbers, then every matrix can be unitarily 'upper triangularized'. Now we take up the question: 'Which matrices can be unitarily diagonalized?'. We saw that a matrix  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is unitarily diagonalizable if and only if there is an orthonormal basis for  $\mathbb{K}^{n \times 1}$  consisting of eigenvectors of  $\mathbf{A}$ . Let us investigate this condition further.

Suppose  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is unitarily diagonalizable, and let eigenvectors  $\mathbf{u}_1, \dots, \mathbf{u}_n$  of  $\mathbf{A}$  form an orthonormal basis for  $\mathbb{K}^{n \times 1}$ . Let  $\lambda_1, \dots, \lambda_n$  be the corresponding eigenvalues of  $\mathbf{A}$ , so that  $\mathbf{A}\mathbf{u}_j = \lambda_j\mathbf{u}_j$  for  $j = 1, \dots, n$ . Let  $\mathbf{x} \in \mathbb{K}^{n \times 1}$ . Then

$$\mathbf{x} = \sum_{j=1}^n \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{u}_j \quad \text{and} \quad \mathbf{A}\mathbf{x} = \sum_{j=1}^n \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{A}\mathbf{u}_j = \sum_{j=1}^n \lambda_j \langle \mathbf{u}_j, \mathbf{x} \rangle \mathbf{u}_j.$$

Thus for any  $\mathbf{x} \in \mathbb{K}^{n \times 1}$ , the column vector  $\mathbf{A}\mathbf{x}$  is completely determined by the inner products  $\langle \mathbf{u}_1, \mathbf{x} \rangle, \dots, \langle \mathbf{u}_n, \mathbf{x} \rangle$ , and by the eigenvalues and eigenvectors of  $\mathbf{A}$ . The above representation of  $\mathbf{A}\mathbf{x}$  can be used for various purposes. For this reason, we would like to find necessary and/or sufficient conditions under which a square matrix can be unitarily diagonalized. We introduce a new class of matrices.

### Definition

A matrix  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is called **normal** if it commutes with its adjoint, that is,  $\mathbf{A}^* \mathbf{A} = \mathbf{A} \mathbf{A}^*$ .

**Examples** (i) If  $\mathbf{A}$  is self-adjoint (i.e.,  $\mathbf{A}^* = \mathbf{A}$ ), or skew self-adjoint (i.e.,  $\mathbf{A}^* = -\mathbf{A}$ ), or unitary, then  $\mathbf{A}$  is normal.

(ii) The matrix  $\mathbf{A} := \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \in \mathbb{K}^{2 \times 2}$  is normal, but it is neither self-adjoint, nor skew self-adjoint, nor unitary. But not every matrix is normal, as the example  $\mathbf{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  shows.

## Proposition

Let  $\mathbf{A} \in \mathbb{K}^{n \times n}$ . Then

$\mathbf{A}$  is normal  $\iff \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle$  for all  $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n \times 1}$ .

Proof. For  $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n \times 1}$ ,  $\langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y} \rangle = (\mathbf{A}\mathbf{x})^* \mathbf{A}\mathbf{y} = \mathbf{x}^* \mathbf{A}^* \mathbf{A}\mathbf{y}$  and  $\langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle = (\mathbf{A}^*\mathbf{x})^* \mathbf{A}^*\mathbf{y} = \mathbf{x}^* \mathbf{A} \mathbf{A}^* \mathbf{y}$ . This shows that if  $\mathbf{A}$  is normal, then  $\langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle$  for all  $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n \times 1}$ .

Conversely, suppose  $\langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^*\mathbf{x}, \mathbf{A}^*\mathbf{y} \rangle$  for all  $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n \times 1}$ . Then letting  $\mathbf{x} := \mathbf{e}_j$  and  $\mathbf{y} := \mathbf{e}_k$ , we see that  $\mathbf{e}_j^* \mathbf{A}^* \mathbf{A} \mathbf{e}_k = \mathbf{e}_j^* \mathbf{A} \mathbf{A}^* \mathbf{e}_k$ , that is, the  $(j, k)$ th entries of  $\mathbf{A}^* \mathbf{A}$  and  $\mathbf{A} \mathbf{A}^*$  are the same for all  $j, k = 1, \dots, n$ . Hence  $\mathbf{A}^* \mathbf{A} = \mathbf{A} \mathbf{A}^*$ , that is,  $\mathbf{A}$  is normal. □

The above condition for normality of a matrix  $\mathbf{A}$  says that the lengths of the vectors  $\mathbf{A}\mathbf{x}$  and  $\mathbf{A}^*\mathbf{x}$  should be the same, and the angle between  $\mathbf{A}\mathbf{x}$  and  $\mathbf{A}\mathbf{y}$  should be the same as the angle between  $\mathbf{A}^*\mathbf{x}$  and  $\mathbf{A}^*\mathbf{y}$  for all  $\mathbf{x}, \mathbf{y} \in \mathbb{K}^{n \times 1}$ .

## Corollary

Let  $\mathbf{A} \in \mathbb{K}^{n \times n}$  be normal. Then  $\|\mathbf{Ax}\| = \|\mathbf{A}^* \mathbf{x}\|$  for all  $\mathbf{x}$  in  $\mathbb{K}^{n \times 1}$ . Let  $\mathbf{x}$  be an eigenvector of  $\mathbf{A}$  corresponding to an eigenvalue  $\lambda$ . Then  $\mathbf{x}$  itself is an eigenvector of  $\mathbf{A}^*$  corresponding to the eigenvalue  $\bar{\lambda}$  of  $\mathbf{A}^*$ . Further, if  $\mathbf{y}$  is an eigenvector of  $\mathbf{A}$  corresponding to an eigenvalue  $\mu \neq \lambda$ , then  $\mathbf{y}$  is orthogonal to  $\mathbf{x}$ .

Proof. Let  $\mathbf{x} \in \mathbb{K}^{n \times 1}$ . Since  $\mathbf{A}$  is normal,

$$\|\mathbf{Ax}\|^2 = \langle \mathbf{Ax}, \mathbf{Ax} \rangle = \langle \mathbf{A}^* \mathbf{x}, \mathbf{A}^* \mathbf{x} \rangle = \|\mathbf{A}^* \mathbf{x}\|^2 \quad \text{for all } \mathbf{x} \in \mathbb{K}^{n \times 1}.$$

Next, let  $\mathbf{x}$  be an eigenvector of  $\mathbf{A}$  corresponding to an eigenvalue  $\lambda$ . Then  $\|\mathbf{A}^* \mathbf{x} - \bar{\lambda} \mathbf{x}\| = \|\mathbf{Ax} - \lambda \mathbf{x}\| = 0$ . Hence  $\mathbf{x}$  itself is an eigenvector of  $\mathbf{A}^*$  corresponding to the eigenvalue  $\bar{\lambda}$ . Finally, let  $\mathbf{y}$  be an eigenvector of  $\mathbf{A}$  corresponding to an eigenvalue  $\mu \neq \lambda$ . Then

$$\mu \langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, \mu \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{Ay} \rangle = \langle \mathbf{A}^* \mathbf{x}, \mathbf{y} \rangle = \langle \bar{\lambda} \mathbf{x}, \mathbf{y} \rangle = \bar{\lambda} \langle \mathbf{x}, \mathbf{y} \rangle.$$

Since  $\mu \neq \bar{\lambda}$ , we see that  $\langle \mathbf{x}, \mathbf{y} \rangle = 0$ , that is,  $\mathbf{x} \perp \mathbf{y}$ . □



We now characterize a diagonal matrix in terms of its upper triangularity and normality.

### Lemma

A matrix  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is diagonal if and only if it is upper triangular and normal.

Proof. Let  $\mathbf{A} := \text{diag}(\lambda_1, \dots, \lambda_n)$ . Then  $\mathbf{A}$  is upper triangular. Also, it is normal since  $\mathbf{A}^* \mathbf{A} = \text{diag}(|\lambda_1|^2, \dots, |\lambda_n|^2) = \mathbf{A} \mathbf{A}^*$ .

Conversely, suppose  $\mathbf{A} := [a_{jk}]$  is upper triangular and normal. Let  $\mathbf{B} := \mathbf{A}^* \mathbf{A} = [b_{jk}]$  and  $\mathbf{C} := \mathbf{A} \mathbf{A}^* = [c_{jk}]$ . Since  $\mathbf{A}$  is upper triangular,  $a_{jk} = 0$  if  $j > k$ , and so for  $k = 1, \dots, n$ ,

$$b_{kk} = \sum_{\ell=1}^n \bar{a}_{\ell k} a_{\ell k} = \sum_{\ell=1}^k |a_{\ell k}|^2 \quad \text{and} \quad c_{kk} = \sum_{\ell=1}^n a_{k\ell} \bar{a}_{k\ell} = \sum_{\ell=1}^n |a_{k\ell}|^2.$$

Since  $\mathbf{A}$  is normal, we see that  $b_{kk} = c_{kk}$  for  $k = 1, \dots, n$ .

Let  $k = 1$ . Then

$$|a_{11}|^2 = b_{11} = c_{11} = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2.$$

Hence  $a_{12} = \cdots = a_{1n} = 0$ .

Next, let  $k = 2$ . Then

$$|a_{22}|^2 = |a_{12}|^2 + |a_{22}|^2 = b_{22} = c_{22} = |a_{22}|^2 + |a_{23}|^2 + \cdots + |a_{2n}|^2.$$

Hence  $a_{23} = \cdots = a_{2n} = 0$ .

Proceeding in this manner, for  $k = n - 1$ , we obtain

$$|a_{(n-1)(n-1)}|^2 = |a_{1(n-1)}|^2 + \cdots + |a_{(n-1)(n-1)}|^2 = b_{(n-1)(n-1)} = c_{(n-1)(n-1)} = |a_{(n-1)(n-1)}|^2 + |a_{(n-1)n}|^2.$$

Hence  $a_{(n-1)n} = 0$ .

Thus  $a_{jk} = 0$  if  $j < k$ , that is,  $\mathbf{A}$  is lower triangular. Since  $\mathbf{A}$  is given to be upper triangular, it is in fact diagonal.  $\square$

We are now ready to state and prove necessary conditions as well as sufficient conditions for diagonalizing a matrix unitarily.

## Proposition (Spectral Theorem for Normal Matrices)

- (i) If  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is unitarily diagonalizable, then  $\mathbf{A}$  is normal.
- (ii) If  $\mathbf{A} \in \mathbb{C}^{n \times n}$  is normal, then  $\mathbf{A}$  is unitarily diagonalizable.

**Proof.** (i) Suppose  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is unitarily diagonalizable. Then  $\mathbf{D} = \mathbf{U}^* \mathbf{A} \mathbf{U}$  for some unitary matrix  $\mathbf{U}$  and diagonal matrix  $\mathbf{D}$ . Hence  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^*$ , and so  $\mathbf{A}^* = \mathbf{U} \mathbf{D}^* \mathbf{U}^*$ . Now

$$\mathbf{A}^* \mathbf{A} = (\mathbf{U} \mathbf{D}^* \mathbf{U}^*)(\mathbf{U} \mathbf{D} \mathbf{U}^*) = \mathbf{U} (\mathbf{D}^* \mathbf{D}) \mathbf{U}^*$$

whereas

$$\mathbf{A} \mathbf{A}^* = (\mathbf{U} \mathbf{D} \mathbf{U}^*)(\mathbf{U} \mathbf{D}^* \mathbf{U}^*) = \mathbf{U} (\mathbf{D} \mathbf{D}^*) \mathbf{U}^*.$$

Since  $\mathbf{D}$  is diagonal,  $\mathbf{D}^* \mathbf{D} = \mathbf{D} \mathbf{D}^*$ . Consequently,  $\mathbf{A}$  is normal.

(ii) Let  $\mathbf{A} \in \mathbb{C}^{n \times n}$ . By Schur's theorem, there is a unitary matrix  $\mathbf{U}$  and an upper triangular matrix  $\mathbf{B}$  with  $\mathbf{B} = \mathbf{U}^* \mathbf{A} \mathbf{U}$ .

Suppose  $\mathbf{A}$  is normal, that is,  $\mathbf{A}^* \mathbf{A} = \mathbf{A} \mathbf{A}^*$ . Then using  $\mathbf{B} = \mathbf{U}^* \mathbf{A} \mathbf{U}$ , we easily see that  $\mathbf{B}^* \mathbf{B} = \mathbf{B} \mathbf{B}^*$ , that is,  $\mathbf{B}$  is normal. Now, since  $\mathbf{B}$  is upper triangular and normal,  $\mathbf{B}$  is diagonal by the previous lemma. This proves that  $\mathbf{A}$  is unitarily diagonalizable. □

### Remarks

(i) The unitary matrix  $\mathbf{U}$  and the diagonal matrix  $\mathbf{B}$  such that  $\mathbf{A} = \mathbf{U} \mathbf{B} \mathbf{U}^*$  are not unique. We shall give some examples later.

(ii) Part (ii) of the proposition is not true for real scalars, that is, even if  $\mathbf{A} \in \mathbb{R}^{n \times n}$  is normal, there may be no unitary  $\mathbf{U} \in \mathbb{R}^{n \times n}$  and diagonal  $\mathbf{D} \in \mathbb{R}^{n \times n}$  such that  $\mathbf{D} = \mathbf{U}^* \mathbf{A} \mathbf{U}$ . For example,  $\mathbf{A} := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$  is normal, but it is not diagonalizable using real scalars since it has no real eigenvalue.

# Self-adjoint matrices

Recall that  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is called **self-adjoint** (or **Hermitian**) if  $\mathbf{A}^* = \mathbf{A}$ . We shall now prove a “spectral theorem” for self-adjoint matrices. For this purpose, the following lemma is useful.

## Lemma

A matrix  $\mathbf{A} \in \mathbb{K}^{n \times n}$  is diagonal with all diagonal entries real if and only if it is upper triangular and self-adjoint.

Proof. Let  $\mathbf{A} := \text{diag}(\lambda_1, \dots, \lambda_n)$  with  $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ . Then clearly  $\mathbf{A}$  is upper triangular. Also, it is self-adjoint since  $\mathbf{A}^* = \text{diag}(\bar{\lambda}_1, \dots, \bar{\lambda}_n) = \text{diag}(\lambda_1, \dots, \lambda_n) = \mathbf{A}$ .

Conversely, suppose  $\mathbf{A}$  is upper triangular and self-adjoint. Then  $\mathbf{A}^*$  is lower triangular. Since  $\mathbf{A}^* = \mathbf{A}$ , we see that  $\mathbf{A}$  is in fact diagonal with all diagonal entries real. □

Just as we have proved that a matrix  $\mathbf{A} \in \mathbb{C}^{n \times n}$  is normal if and only if it is unitarily diagonalizable, we prove a similar result for self-adjoint matrices.

### Proposition (Spectral Theorem for Self-Adjoint Matrices)

Let  $\mathbf{A} \in \mathbb{C}^{n \times n}$ . Then  $\mathbf{A}$  is self-adjoint if and only if  $\mathbf{A}$  is unitarily diagonalizable and all eigenvalues of  $\mathbf{A}$  are real.

Proof. Suppose  $\mathbf{A}$  is unitarily diagonalizable and all eigenvalues of  $\mathbf{A}$  are real. Let  $\mathbf{U}$  be a unitary matrix and let  $\mathbf{D}$  be a diagonal matrix such that  $\mathbf{D} = \mathbf{U}^* \mathbf{A} \mathbf{U}$ . Then the diagonal entries of  $\mathbf{D}$  are the eigenvalues of  $\mathbf{A}$ , and so they are real. Hence  $\mathbf{D}^* = \mathbf{D}$ . Consequently,

$$\mathbf{A}^* = (\mathbf{U} \mathbf{D} \mathbf{U}^*)^* = \mathbf{U} \mathbf{D}^* \mathbf{U}^* = \mathbf{U} \mathbf{D} \mathbf{U}^* = \mathbf{A}.$$

Thus  $\mathbf{A}$  is self-adjoint.

Conversely, suppose  $\mathbf{A}$  is self-adjoint. By Schur's theorem, there is a unitary matrix  $\mathbf{U}$ , and also an upper triangular matrix  $\mathbf{B}$  such that  $\mathbf{B} = \mathbf{U}^* \mathbf{A} \mathbf{U}$ . Then

$$\mathbf{B}^* = (\mathbf{U}^* \mathbf{A} \mathbf{U})^* = \mathbf{U}^* \mathbf{A}^* \mathbf{U} = \mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{B},$$

so that  $\mathbf{B}$  is self-adjoint. Since  $\mathbf{B}$  is upper triangular and self-adjoint, the previous lemma shows that  $\mathbf{B}$  is in fact diagonal with all diagonal entries real. Thus  $\mathbf{A}$  is unitarily diagonalizable. Also, all eigenvalues of  $\mathbf{A}$  are real, since they are the diagonal entries of the matrix  $\mathbf{B}$ . □

Our short proof of the spectral theorem for self-adjoint matrices is based on Schur's theorem. This result can also be deduced from part (ii) of the spectral theorem for normal matrices since every self-adjoint matrix is normal, provided we independently show that every eigenvalue of a self-adjoint matrix is real. The latter statement can be easily proved.

## Proposition

If  $\mathbf{A} \in \mathbb{C}^{n \times n}$  is self-adjoint, then every eigenvalue of  $\mathbf{A}$  is real.

Proof.

Let  $\lambda$  be an eigenvalue of a self-adjoint matrix  $\mathbf{A}$ , and let  $\mathbf{x}$  be a corresponding unit eigenvector. Then

$$\lambda = \lambda \mathbf{x}^* \mathbf{x} = \mathbf{x}^* \mathbf{A} \mathbf{x} = \mathbf{x}^* \mathbf{A}^* \mathbf{x} = (\mathbf{A} \mathbf{x})^* \mathbf{x} = (\lambda \mathbf{x})^* \mathbf{x} = \bar{\lambda} \mathbf{x}^* \mathbf{x} = \bar{\lambda}.$$

Hence  $\lambda$  is real. □

Finally, let us consider a real symmetric matrix  $\mathbf{A}$ , that is,  $\mathbf{A} \in \mathbb{R}^{n \times n}$  such that  $\mathbf{A}^T = \mathbf{A}$ . We shall prove a spectral theorem for  $\mathbf{A}$  which involves only real scalars.

A unitary matrix with real entries is also called an **orthogonal matrix**. Thus  $\mathbf{C} \in \mathbb{R}^{n \times n}$  is orthogonal if its columns form an orthonormal subset of  $\mathbb{R}^{n \times 1}$ . Clearly, an orthogonal matrix is invertible and its inverse is the same as its transpose.



A matrix  $\mathbf{A} \in \mathbb{R}^{n \times n}$  is called **orthogonally diagonalizable** if there is an orthogonal matrix  $\mathbf{C} \in \mathbb{R}^{n \times n}$  and a diagonal matrix  $\mathbf{D} \in \mathbb{R}^{n \times n}$  such that  $\mathbf{D} = \mathbf{C}^{-1}\mathbf{A}\mathbf{C}$ . We prove [Jacobi's theorem](#).

### Proposition (Spectral Theorem for Real Symmetric Matrices)

Let  $\mathbf{A} \in \mathbb{R}^{n \times n}$ . Then  $\mathbf{A}$  is symmetric if and only if  $\mathbf{A}$  is orthogonally diagonalizable. In this case,  $\mathbf{A}$  has  $n$  real eigenvalues counted according to their algebraic multiplicities.

Proof.

Suppose  $\mathbf{A}$  is orthogonally diagonalizable. Let  $\mathbf{C} \in \mathbb{R}^{n \times n}$  be an orthogonal matrix, and let  $\mathbf{D} \in \mathbb{R}^{n \times n}$  be a diagonal matrix such that  $\mathbf{D} = \mathbf{C}^T \mathbf{A} \mathbf{C}$ . Since  $\mathbf{D}^T = \mathbf{D}$ ,

$$\mathbf{A}^T = (\mathbf{C} \mathbf{D} \mathbf{C}^T)^T = \mathbf{C} \mathbf{D}^T \mathbf{C}^T = \mathbf{C} \mathbf{D} \mathbf{C}^T = \mathbf{A}.$$

Thus  $\mathbf{A}$  is symmetric.

Conversely, suppose  $\mathbf{A}$  is symmetric. Since  $\mathbb{R}$  can be considered as a subset of  $\mathbb{C}$ , we treat  $\mathbf{A}$  as a matrix with complex entries. Then  $\mathbf{A}^* = \mathbf{A}$ , that is,  $\mathbf{A}$  is self-adjoint, and so all eigenvalues of  $\mathbf{A}$  are real.

By the spectral theorem for self-adjoint matrices, there is a unitary matrix  $\mathbf{U} \in \mathbb{C}^{n \times n}$  and a diagonal matrix  $\mathbf{D} \in \mathbb{C}^{n \times n}$  such that  $\mathbf{D} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$ , that is,  $\mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{D}$ . Since the diagonal entries of  $\mathbf{D}$  are the eigenvalues of  $\mathbf{A}$ , we see that  $\mathbf{D} \in \mathbb{R}^{n \times n}$ .

The  $n$  columns of the unitary matrix  $\mathbf{U}$  form an orthonormal set in  $\mathbb{C}^{n \times n}$ , and each column is an eigenvector of  $\mathbf{A}$  corresponding to an eigenvalue of  $\mathbf{A}$ .

Let  $\lambda$  be an eigenvalue of  $\mathbf{A}$ . Then  $\lambda \in \mathbb{R}$ . Using the [Gauss Elimination Method](#), we may find the basic solutions of the homogeneous linear system  $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ . Their entries are real since all entries of  $\mathbf{A}$  are real and  $\lambda \in \mathbb{R}$ . These solutions form a basis for the eigenspace of  $\mathbf{A}$  corresponding to  $\lambda$ .

Further, we can use the [Gram-Schmidt Orthogonalization Process](#) for these basic solutions to obtain an orthonormal basis for the eigenspace of  $\mathbf{A}$  corresponding to  $\lambda$ . In this process, the entries of the basis vectors remain real.

We replace the  $n$  columns of the unitary matrix  $\mathbf{U}$  by  $n$  eigenvectors of  $\mathbf{A}$  which form an orthonormal set in  $\mathbb{R}^{n \times n}$ . Let  $\mathbf{C} \in \mathbb{R}^{n \times n}$  denote the matrix with these columns arranged in the same order as the columns of  $\mathbf{U}$ . Then  $\mathbf{D} = \mathbf{C}^{-1}\mathbf{A}\mathbf{C}$ .  $\square$

The spectral theorem says that given a normal matrix  $\mathbf{A} \in \mathbb{C}^{n \times n}$  or a symmetric matrix  $\mathbf{A} \in \mathbb{R}^{n \times n}$ , there is a unitary matrix  $\mathbf{U}$  and a diagonal matrix  $\mathbf{D}$  such that  $\mathbf{D} = \mathbf{U}^{-1}\mathbf{A}\mathbf{U}$ , that is,  $\mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{D}$ . Let us write the matrix  $\mathbf{U}$  in terms of its  $n$  columns  $\mathbf{U} = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$  and let  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ .

Then the equation

$$\mathbf{A} \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} \text{diag}(\lambda_1, \dots, \lambda_n)$$

means  $\mathbf{A}\mathbf{u}_1 = \lambda_1\mathbf{u}_1, \dots, \mathbf{A}\mathbf{u}_n = \lambda_n\mathbf{u}_n$ , so that  $\mathbf{u}_1, \dots, \mathbf{u}_n$  are eigenvectors of  $\mathbf{A}$  corresponding to the eigenvalues  $\lambda_1, \dots, \lambda_n$  respectively. We may list the eigenvectors  $\mathbf{u}_1, \dots, \mathbf{u}_n$  in some other order to form another unitary matrix and correspondingly change the ordering of the eigenvalues  $\lambda_1, \dots, \lambda_n$  to form another diagonal matrix which will serve the same purpose.

Since the eigenvalues  $\lambda_1, \dots, \lambda_n$  of  $\mathbf{A}$  may not be distinct, we may pool together all eigenvectors among  $\mathbf{u}_1, \dots, \mathbf{u}_n$  corresponding to the same eigenvalue.

Also, since  $\mathbf{A}$  is diagonalizable, the geometric multiplicity of any eigenvalue of  $\mathbf{A}$  is equal to its algebraic multiplicity.

Thus if we know the eigenvalues of  $\mathbf{A}$ , then we may use the following procedure to form a unitary matrix  $\mathbf{U}$  whose  $n$  columns are eigenvectors of  $\mathbf{A}$ .

1. Let  $\mu_1, \dots, \mu_k$  be the distinct eigenvalues of  $\mathbf{A}$ . These are the distinct roots of the characteristic polynomial of  $\mathbf{A}$ . Let the algebraic multiplicity of  $\mu_j$  be  $m_j$ , so that  $m_1 + \dots + m_k = n$ . Also, the geometric multiplicity  $g_j$  of  $\mu_j$  is equal to  $m_j$ , and so  $g_1 + \dots + g_k = n$ .

If in fact,  $\mathbf{A}$  is self-adjoint, then each  $\mu_j$  is real.

2. For each  $j = 1, \dots, k$ , find a basis for the null space  $\mathcal{N}(\mathbf{A} - \mu_j \mathbf{I})$  consisting of  $g_j$  elements by solving the homogeneous linear system  $(\mathbf{A} - \mu_j \mathbf{I})\mathbf{x} = \mathbf{0}$  using the Gauss Elimination Method.

3. For each  $j = 1, \dots, k$ , obtain an ordered orthonormal basis  $(\mathbf{u}_{j,1}, \dots, \mathbf{u}_{j,g_j})$  for  $\mathcal{N}(\mathbf{A} - \mu_j \mathbf{I})$  using the Gram-Schmidt Orthonormalization Process.

4. Form an  $n \times n$  matrix  $\mathbf{U}$  as follows:

$$\mathbf{U} := \begin{bmatrix} \mathbf{u}_{11} & \dots & \mathbf{u}_{1g_1} & \mathbf{u}_{21} & \dots & \mathbf{u}_{2g_2} & \dots & \dots & \mathbf{u}_{k1} & \dots & \mathbf{u}_{kg_k} \end{bmatrix}$$

Now since  $\mathbf{A}$  is a normal matrix,  $\mathbf{u}_{ik} \perp \mathbf{u}_{j\ell}$  if  $i \neq j$ . Thus the  $n$  columns of  $\mathbf{U}$  form an orthonormal set. Hence  $\mathbf{U}$  is unitary.

Form an  $n \times n$  diagonal matrix  $\mathbf{D}$  as follows:

$$\mathbf{D} := \text{diag}(\lambda_{11}, \dots, \lambda_{1g_1}, \lambda_{21}, \dots, \lambda_{2g_2}, \dots, \dots, \lambda_{k1}, \dots, \lambda_{kg_k}),$$

where  $\lambda_{11} = \dots = \lambda_{1g_1} = \mu_1, \dots, \lambda_{k1} = \dots = \lambda_{kg_k} = \mu_k$ .

Then  $\mathbf{U}^* \mathbf{A} \mathbf{U} = \mathbf{D}$ .

A similar procedure works for a real symmetric matrix  $\mathbf{A}$ , and so we can find an orthogonal matrix  $\mathbf{C} \in \mathbb{R}^{n \times n}$  and a diagonal matrix  $\mathbf{D} \in \mathbb{R}^{n \times n}$  such that  $\mathbf{C}^T \mathbf{A} \mathbf{C} = \mathbf{D}$ .